MIL-B-52207B
7 July 1976
SUPERSEDING
MIL-B-52207A
2 April 1969

MILITARY SPECIFICATION

BOOM, JIB, CRANE, 15-FOOT-LONG, FOR 3/4-CUBIC-YARD (12-1/2-TON CRAWLER MOUNTED, 20-TON WHEEL MOUNTED) CRANES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a 15-foot jib boom to be mounted and used on a military boom conforming to MIL-B-52203.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

GG-P-455	- Plates and Foils, Photographic (Photosensitive Anodized Aluminum).

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-3580	- Cranes and Crane-Shovels, Truck, Crawler, and Wheel Mounted, Full Revolving, and Their Attachments, Packaging of.

FSC 3815

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MIL-G-10924 - Grease, Automotive and Artillery.
MIL-B-52203 - Boom, Crane: 20-Ton, Rough Terrain; 20-Ton, Truck-Mounted; 12-1/2-Ton, Crawler-Mounted.

DRAWINGS

ME
TA13217E3080 - Boom, Jib, Crane, 15 Feet Long.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IV, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, 2501 NW 7th Street, Miami FL 33125.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The jib boom shall consist of a 15-foot main boom section with point assembly, strut frame assembly, and yoke assembly with the necessary riser and holdback ropes. The jib boom shall be capable of lifting 7500 pounds when mounted on the outer boom section of a crane boom conforming to MIL-B-52203. The jib boom shall be as shown on Top Assembly TA13217E3080 and as specified herein.

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3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish a jib boom for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce jib booms that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on applicable drawings.

3.4 Overhaul ball. Unless otherwise specified (see 6.2), an overhaul ball with a forged steel swivel hook shall be furnished with each jib boom. The ball shall be adaptable to the open wedge socket cable fitting, and provided with sealed roller-type thrust bearings in the swivel. The ball and hook assembly shall weigh not less than 125 pounds and have a rated lifting capacity of not less than 7500 pounds.

3.5 Delivery lubrication service. When delivered, all components requiring lubrication, except those for which preservatives are specified in 5.1, shall be lubricated. Grease lubrication shall be with greases conforming to MIL-G-10924.

3.6 Identification marking. The identification marking shall be applied to the jib boom on plates conforming to GG-P-455, Type I, Grade A, Class 1 material and shall include the following information:

FSN _____*
BOOM , JIB, CRANE; 15 FT 12-1/2 TON CRLR, MTD, AND 20 TON TRK. MTD.
Applicable publications _____*
Mfr _____*
Model _____* Contract No. _____*
Serial No. _____*
Capacity 7500 lbs
Weight _____
FOR USE WITH MIL. STD. BOOM FOR 12-1/2 TON CRLR. MTD. AND 20 TON TRK. MTD.
*Contractor to fill in the blanks.

Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

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3.7 Treatment and painting. The jib boom, its components, and parts normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, color as specified (see 6.2).

3.8 Technical publications. Such technical publications as are specified (see 6.2) shall be furnished.

3.9 Workmanship. The jib boom shall be free from defects such as incomplete welds, cracks, sharp edges, rust, or other defects that could impair the operation or serviceability of the jib boom. Boltholes shall be accurately punched or drilled and shall have all burrs removed. All surfaces to be adjoined shall be correctly alined.

3.9.1 Welders and welding.

3.9.1.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualification.

Contractors who only make horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welder equipment operators are capable of producing quality welds. Operators of any automatic, semi-automatic or manual arc-welding machines shall be considered welders for the purpose of this specification.

3.9.1.2 Welding. Surfaces of steel parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when parts connected by welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the bucket to conform to examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall

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be brought into as close contact as possible and in no event shall be separated by more than 1/16 inch. The welding process, and the procedures to be followed to fabricate the boom shall conform to the requirements of AWS D1.1 with revisions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performanceof all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor, is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.3 Preproduction inspection. Prior to examination or test, the preproduction jib boom shall be completely lubricated with the grease specified in 3.5.

4.3.1 Examination. The preproduction jib boom shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.3.2 Test. The preproduction jib boom shall be tested as specified in 4.5.2. Failure of the test shall be cause for performing the inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Examination. Each jib boom shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The jib boom shall be examined as specified herein for the following major defects:

101. Mounting holes in jib boom foot not as specified.
102. Sheave pin bushing bore of jib boom point not as specified.
103. Other dimensions not as specified.
104. Parts or components not as specified.
105. Material not as specified.
106. Poor fit at joints, seams, or edges, or misalinement.
107. Workmanship not as specified.
108. Treatment and painting not as specified.
109. Lubrication fittings not as specified.
110. Identification or other marking not as specified.
111. Assembly not complete or not correct.

4.5.2 Weld and structural test. The jib boom shall be assembled in a suitable structure to simulate attachment for operation of the jib boom to a crane boom conforming to MIL-B-52203. The assembly of the jib boom must utilize all components of the jib boom including the strut assembly and necessary holdback ropes. The jib boom shall be set so that the longitudinal axis of the jib boom is 5 degrees above the horizontal. A 7500-pound load shall be applied to the load line, or by hoisting on a fixed anchor equipped with a force measuring device. Then a side load of 150 pounds shall be applied once from each direction. The side load can be applied at either the suspended load or at the jib boom point. The applied load shall be released and reapplied with successive side loads for a total of 10 cycles. Inability to assemble the jib boom, misalinement or binding of any sheaving, any evidence of breakage, permanent deformation, or weld failure shall constitute failure of this test.

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4.6 Inspection of packaging. The preservation, packing, and marking shall be examined to determine compliance with the quality assurance provisions of MIL-C-3580.

5. PACKAGING

5.1 Preservation, packing, and marking. The jib boom, overhaul ball, and technical publications shall be preserved, packed, and marked in accordance with MIL-C-3580. The preservation and packing shall be Level A, B, or Commercial, as specified (see 6.2).

6. NOTES

6.1 Intended use. The jib boom is intended for use on 3/4-cubic-yard-size (12-1/2-ton crawler-mounted and 20-ton truck-mounted) cranes equipped with a crane boom conforming to MIL-B-52203.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction model (see 3.2).
(c) When an overhaul ball is not required (see 3.4).
(d) Color required (see 3.7).
(e) Technical publications required (see 3.8).
(f) Degree of preservation and packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production jib booms from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish jib booms conforming to this specification.

Custodians:
Army - ME

Preparing activity:
Army - ME

Review activities:
Army - CE
DSA - CS

Project 3815-0083

User activity:
Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-B-52207B BOOM, JIB, CRANE, 15-FOOT-LONG, FOR, 3/4-CUBIC-YARD (12-1/2-TON CRAWLER MOUNTED, 20-TON WHEEL MOUNTED) CRANES

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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